



Experiencias de aprendizaje 06

Ejercicios - Los ejercicios deben resolverse a mano

1. $y = -18$ $y' = 0$

2. $y = x^{6-5}$ $y = 6x^5$

3. $y = x^{9-1}$ $y' = 9x^8$

4. $y = 7x^{3-1} - 4x^{1-1}$ $y' = 21x^2 - 4$

5. $y = 6x^{3-1} + 3x^{2-1} - 10^0$ $y' = 18x^2 + 6x$

6. $y = 4\sqrt{x} - \frac{6}{\sqrt[3]{x^2}}$

$y = 4x^{1/2} - \frac{6}{x^{2/3}}$

$y = 4x^{1/2} - 6x^{-2/3}$

$y' = 4\left(\frac{1}{2}\right)x^{-1/2} - 6\left(-\frac{2}{3}\right)x^{-5/3}$

$y' = 2x^{-1/2} + 4x^{-5/3}$

$y' = \frac{2}{\sqrt{x}} + \frac{4}{\sqrt[3]{x^5}}$

7. $y = \frac{x-x^2}{\sqrt{x}}$ $y' = \frac{1}{2}x^{-1/2} - \frac{3}{2}x^{1/2}$

$y = \frac{x-x^2}{x^{1/2}}$ $y' = \frac{1}{2}x^{-1/2} - \frac{3}{2}x^{1/2}$

$y = (x-x^2)(x^{-1/2})$ $y' = \frac{1}{2\sqrt{x}} - \frac{3}{2}\sqrt{x}$

$y = x^{1/2} - x^{3/2}$

8. $f(x) = \frac{1}{5}x^{5-1} - 3x^{4-1} + 9x^{2-1} + 1$ $f'(x) = 1x^4 - 12x^3 + 18x$

9. $f(x) = \frac{2}{3}x^{6-1} + 4x^{3-1} - 13x^{2-1} + 8x^{1-1} + 2$

$f'(x) = 4x^5 + 12x^2 - 26x + 8$



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10. $y = \frac{10}{x^2+1}$

$$y' = \frac{(x^2+1)(0) - 10(2x)}{(x^2+1)^2}$$

$$y' = \frac{-20x}{(x^2+1)^2}$$

11. $y = \frac{5}{4x-3}$

$$y' = \frac{(4x-3)(0) - 5(4)}{(4x-3)^2}$$

$$y' = \frac{-20}{(4x-3)^2}$$

12. $y = \frac{3x+1}{2x-5}$

$$y' = \frac{(2x-5)(3) - (3x+1)(2)}{(2x-5)^2}$$

$$y' = \frac{6x-15 - (6x+2)}{(2x-5)^2}$$

$$y' = \frac{6x-15-6x-2}{(2x-5)^2} = \frac{-17}{(2x-5)^2}$$

13. $y = \frac{2-3x}{7-x}$

$$y' = \frac{(7-x)(-3) - (2-3x)(-1)}{(7-x)^2}$$

$$y' = \frac{-21+3x - (-2+3x)}{(7-x)^2}$$

$$y' = \frac{-21+3x+2-3x}{(7-x)^2} = \frac{-19}{(7-x)^2}$$



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14. $y = x^2 - \cos x$

$$y' = 2x - (-\sin x)$$

$$y' = 2x + \sin x$$

15. $y = 4x^{3-1} + x + 5\sin x$

$$y' = 12x^2 + 1 + 5\cos x$$

16. $y = x \sin x$

$$y' = \overset{(1)}{x} \cos x + \sin x \overset{(1)}{(1)}$$

$$y' = x \cos x + \sin x$$

17. $y = (x^3 - 2)\sin x$

$$y' = (x^3 - 2)\cos x + \sin x (3x^2)$$

$$y' = (x^3 - 2)\cos x + 3x^2 \sin x$$

18. $y = \cos x \cot x$

$$y' = \cos x (-\csc^2 x) + \cot x (-\sin x)$$

$$y' = -\cos x \cdot \csc^2 x - \cot x \cdot \sin x$$



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19. $f(x) = \frac{2}{\cos x \cot x}$

$y' =$

$$f'(x) = \frac{(\cos x \cot x)' (2) - 2(-\cos x \cdot \csc^2 x - \cot x \cdot \sin x)}{(\cos x \cot x)^2}$$

$$f'(x) = \frac{-2(-\cos x \csc^2 x - \cot x \cdot \sin x)}{(\cos x \cdot \cot x)^2}$$



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20. $f(x) = \frac{2+\sin x}{x}$

$$f'(x) = \frac{x(\cos x) - (2+\sin x)(1)}{x^2}$$

$$f'(x) = \frac{x \cos x - 2 - \sin x}{x^2}$$